

Matthew Preisser

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EDUCATION

University of Texas at Austin

Aug. 2018 – May 2025

- PhD, Civil Engineering: March 2025 Defense, May 2025 Graduation
- *Dissertation: Description, prediction, & prescription of compounding hazards on society*
- Master's Degree, Environmental and Water Resource Engineering: Dec. 2021
- Master's Degree, Public Affairs: Dec. 2021
- *Thesis: Intersecting near real-time fluvial and pluvial inundation estimates with sociodemographic vulnerability to quantify a household flood impact index*

Auburn University

Aug. 2014 – May 2018

- Bachelor's Degree, Biosystems Engineering, minors in Sustainability Studies & German: May 2018
- *Thesis: Anaerobic digestion of food waste and poultry litter for biogas production*
- Alabama Society of Professional Engineers Outstanding Student of the Year (2018)
- Auburn University Biosystems Engineering Outstanding Student of the Year (2018)

PROFESSIONAL APPOINTMENTS

Postdoctoral Researcher

August 1st Start

Department of Civil, Environmental and Geomatic Engineering

ETH Zurich

- Advisor: Dr. Paola Passalacqua

Graduate Research Assistant

Aug. 2018 – May 2015

Maseeh Dept. of Civil, Architectural and Environmental Engineering *University of Texas at Austin*

- Advisor: Dr. Paola Passalacqua
- Committee: Dr. R. Patrick Bixler, Dr. Matt Bartos, & Dr. Stephen Boyles
- Primary Funding: NSF GRFP, NASA FINESST
- Research Collaborations, DoE SETx-IUFL, PlanetTexas2050, & NASA ATX Climate Atlas

GIS Research Analyst

Jun. 2018 – Aug. 2018

NASA DEVELOP

Marshall Space Flight Center

- Advisors: Dr. Jeffrey Luvall & Dr. Robert Griffin
- *Assessing Water Quality in Thailand's Chao Phraya Watershed by Modeling Sediment Conc.*
- Primary Funding: NASA DEVELOP and SSAI

Undergraduate Research Fellow

Jan. 2017 – May 2018

Department of Biosystems Engineering

Auburn University

- Advisor: Dr. Brendan Higgins
- *Waste products for algae and biogas and sustainable energy production*
- Primary Funding: Auburn University Honors College

UNDER REVIEW AND IN PREPARATION PUBLICATIONS

8. **Preisser, M.**, P. Passalacqua, (2025), Inequality in Socio-Hydrology: The Role of Adaptive Capacity in Urban Flood Resilience. *In Preparation.*
7. Rosenheim, N., L. Mezei, **M. Preisser**, C. Brelsford, P. Bixler, M. Meyer, F. Ahmed (2024), Social Vulnerability Indices: Exploring Assumptions and Limitations. *Global Environmental Change. Under Review.*

PEER REVIEWED PUBLICATIONS

6. **Preisser, M.**, P. Passalacqua, (2025), Remote Sensing Improves Multi-Hazard Flooding and Extreme Heat Detection by Fivefold over Current Estimates. *AGU Advances*, doi: [10.1029/2025AV001667](https://doi.org/10.1029/2025AV001667)
5. **Preisser, M.**, P. Passalacqua, S. Boyles, P. Bixler, (2023), A network-based analysis of critical resource accessibility during flooding, *Frontiers in Water*, doi: [10.3389/frwa.2023.1278205](https://doi.org/10.3389/frwa.2023.1278205)
4. Bixler, R. P., S. Paul, D. Bhakta, T. Farchy, J. Olson, **M. Preisser**, and P. Passalacqua, (2023), 15. Adaptive governance for disaster risk reduction, *Handbook on Adaptive Governance*, 233-251 ISBN: 9781800888234
3. **Preisser, M.**, P. Passalacqua, P. Bixler, J. Hofmann. (2022), Intersecting near-real time fluvial and pluvial inundation estimates with sociodemographic vulnerability to quantify a household flood impact index, *Hydrology and Earth System Sciences*, 26, 3941-3964 doi: [10.5194/hess-26-3941-2022](https://doi.org/10.5194/hess-26-3941-2022)
2. RP. Bixler, S. Paul, J. Jones, **M. Preisser**, P. Passalacqua, (2021), Unpacking adaptive capacity to flooding in urban environments: social capital, social vulnerability, and risk perception, *Frontiers in Water*, 101, doi: [10.3389/frwa.2021.728730](https://doi.org/10.3389/frwa.2021.728730)
1. Chaump, K., **M. Preisser**, S. Shanmugam, R. Prasad, S. Adhikari, B. Higgins (2019), Leaching and anaerobic digestion of poultry litter for biogas production and nutrient transformation, *Waste Management*, 84, 413-422, doi: [10.1016/j.wasman.2018.11.024](https://doi.org/10.1016/j.wasman.2018.11.024)

SOFTWARE PACKAGES

SVInsight | *Open-source tool for estimating social vulnerability indices*

[Github: v1.1.2](#)

pygeoflood | *Near-real-time tool for estimating fluvial, pluvial, and coastal flooding*

[Github: Beta](#)

GRANTS AND FELLOWSHIPS

Total Grants and Fellowships Received (2018-Present): \$274,750

7. Consortium of Universities for the Advancement of Hydrologic Science Hydroinformatics Fellow
 - *Building out a near-real-time compound inundation python package*
 - Team: Dr. Paola Passalacqua and Mark Wang
 - Obligated Funding: \$5,000 total, \$2,500 to Preisser
 - Period: April 2024 – Feb. 2025
6. NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellow
 - *Application of Earth Observation Data in Multi-Layer Network Analysis of Flood Hazards, Social Vulnerability, and Risk*
 - Advisor: Dr. Paola Passalacqua
 - Obligated Funding: \$100,000
 - Period: Aug. 2022 – Sept. 2024
5. National Science Foundation Graduate Research Internship Program (GRIP) Fellow
 - *USGS National Land Imaging Program Public Affairs and Science Communication Internship*
 - Mentor: Timothy Stryker
 - Obligated Funding: \$5,000

- Period: Sept. 2019 – Aug. 2022
4. National Science Foundation Graduate Research Fellow (NSF-GRFP)
 - *Incorporating flood mapping with social disparities to identify at risk communities during extreme weather events*
 - Advisor: Dr. Paola Passalacqua
 - Obligated Funding: \$150,000
 - Period: May 2020 – Aug. 2020
 3. Strauss Center's Brumley Next Generation Fellowship
 - *Visualizing climate change threats in relationship to power generation capabilities in India*
 - Mentor: Dr. Joshua Busby
 - Obligated Funding: \$5,000
 - Period: Sept. 2019 – May 2020
 2. Supplemental Professional Development and Travel Grants
 - UT Civil, Architectural, and Environmental Engineering Kolodzey Travel Grant: \$1,000
 - UT Division of Diversity and Engagement Professional Development Award (2023): \$250
 - UT Dean's Prestigious Supplement Fellowship (2019-2023): \$11,000
 1. Pre-2018 Academic Scholarships
 - Birdsong Endowed Engineering Scholarship (2017)
 - Fred A. Kummer Endowed Scholarship (2015)
 - Hop King Endowed Scholarship (2015)
 - Auburn University Presidential Academic Scholarship (2014-2018)

HONORS AND AWARDS

- Civil Engineering Symposium 2nd Place Poster (2021)
- Patagonia Sustainability Case Competition International Top 10 Finalist (2019)
- Auburn University Biosystems Engineering Outstanding Student of the Year (2018)
- Alabama Society of Professional Engineers Outstanding Student of the Year (2018)
- Inducted into Phi Beta Kappa Society (2018)
- Mark A. Spencer Creative Mentorship Award (2018)
- Semi-Finalist for the Fulbright Scholarship (2018)
- ReNUWIt Best Overall Research Poster (2017)
- Nominated for the Truman Scholarship (2017)
- Nominated for the Udall Scholarship (2017)
- Inducted into Tau Beta Pi Honor Society (2016)

PROFESSIONAL SERVICE AND VOLUNTEERING

Journal Reviews

Frontiers in Water, Hydrological Processes, Journal of Infrastructure Systems

AGU Council

Elected representative, advising leadership committee on scientific affairs

Jan. 2023 – Present

AGU H3S Committee

Treasurer of Hydrology Section Student Subcommittee

Jan. 2022 – Dec. 2024

AGU Thriving Earth Exchange Community Science Fellowship

Coordinated a team of volunteers to address flood and lead related issues

Mar. 2021 – Mar. 2022

NASA DEVELOP Ambassador

Program ambassador corps representative

Aug. 2018 – May 2019

MENTORING EXPERIENCE

UT Austin Environmental Science Institute Mentor

Supervised an undergraduate REU student on a summer project

May 2024 – Aug. 2024

NASA ATX Climate Atlas Mentor

Supervised an undergraduate student on a flood mapping project

Sept. 2023 – May 2024

INVITED PRESENTATIONS

2. Passalacqua, P., **M. Preisser**, M. Wang (2024), Acute on Chronic Stressors in Communities: When Flooding Is Not the Only Issue, NH23A-2272 Invited talk presented at 2024 Fall Meeting, AGU, 9-13 December.
1. Passalacqua, P., **M. Preisser**, M. Wang, P. Bixler, A. Hooks, J. Hofmann, L. Haselbach, H. Moftakhari, H. Evans, C. Thies, D. Maidment (2022), Preparing for Future Floods: Leveraging Remotely Sensed Data, Modeling, and Social Science Information in a Multilayer Network Approach, H46D-01 Invited talk presented at 2022 Fall Meeting, AGU, 12-16 December.

PRESENTATIONS

12. **Preisser, M.**, P. Passalacqua (2024), Missing Hazards: Improving detection and delineation methods of multi-hazard events and assessing their impact on vulnerable populations, GC31V-0118 presented at 2024 Fall Meeting, AGU, 9-13 December.
11. Wang, M., **M. Preisser**, P. Passalacqua (2024), A Near Real-Time Compound Flood Inundation Python Package, NH41D-2342 presented at 2024 Fall Meeting, AGU, 9-13 December.
10. **Preisser, M.**, P. Passalacqua, P. Bixler, H. Kamath, D. Niyogi (2023), Multiple hazards from multiple satellites: Analyzing exposure to cascading hazards using daily Earth observation data from the past decade, NH33D-0822 presented at the 2023 Fall Meeting, AGU, 11-15 December.
9. **Preisser, M.**, P. Passalacqua, P. Bixler, (2023), A network-based disaster resilience metric for estimating individuals' loss of access to critical resources during flooding, presented at 2023 Annual Meeting, EGU, 23-28 April.
8. **Preisser, M.**, P. Passalacqua, P. Bixler (2022), Helping Those Who Need It Most: Open Access Tool for Modeling Flood Hazards and Community Adaptive Capacity in Near-Real Time, H55M-0745 presented at 2022 Fall Meeting, AGU, 12-16 December.
7. Karimaghaei, S., P. Passalacqua, **M. Preisser**, P. Bixler, (2022) Identifying Morphological Hot Spots and Social Vulnerabilities along the Gulf Coast using Remote Sensing, EP15B-1087 presented at 2022 Fall Meeting, AGU, 12-16 December.
6. **Preisser, M.**, P. Passalacqua, P. Bixler (2021), Interconnecting Networks of Social Vulnerability, Resource Access, and High-Resolution Inundation to Quantify Household Flood Impact, H55V-0993 presented at 2021 Fall Meeting, AGU, 13-17 December.
5. **Preisser, M.**, P. Passalacqua, P. Bixler (2020), A Multiplex approach to integrate social vulnerability into urban flood mapping, H139-0015A presented at 2020 Fall Meeting, AGU, 1-17 December.

4. **Preisser, M.**, N. Molder, K. Benedict, C. Shapiro, C. Straub, T. Stryker (2020), Expanding the Societal Benefits of Earth Observations through Leading Practices for Public Private Partnerships, SY003-0002 presented at 2020 Fall Meeting, AGU, 1-17 December.
3. Newman, T., Z. Wu, B. Reed, T. Stryker, **M. Preisser** (2020), Assessing COVID-19 environmental impacts through remote sensing, U012-05 presented at 2020 Fall Meeting, AGU, 1-17 December.
2. Garcia, K., **M. Preisser**, C. Nauman (2019), Assessing Water Quality in Thailand's Chao Phraya Watershed by Modeling Sediment Concentration and Urban Footprint, presented at 2019 Annual Meeting, AAG, 5 April, & 2018 Wernher von Braun Memorial Symposium, AAS, 23-25 October.
1. **Preisser, M.**, B. Higgins (2018), Anaerobic digestion of poultry litter and food waste for biogas production, presented at National Conference on Undergraduate Research, NCUR, 4-7 April.

CONVENED SESSIONS AND TOWN HALLS

5. Miralha L., D. Peterson, **M. Preisser**, A. Akerele, Y. Che, J. Lerback, (2024), Intersections in Science Roundtable: Building Cross-Disciplinary Connections for Early Career Researchers, Co-Convened Town Hall at 2024 Annual Meeting, AGU, 9-13 December.
4. Ellis, E., P. Becker, A. Price, **M. Preisser**, (2023), Communicating Science Beyond the Paper: Thinking Outside the Boxplot, INV13A Co-Convened Innovative Session at 2023 Annual Meeting, AGU, 11-15 December.
3. Peterson, D., L. Miralha, J. Lerbeck, **M. Preisser**, J. Allen, (2023), Where and How Can AGU Assist? Discussion About the Future of Early-Career and Student Members Within AGU, TH33G Co-Convened Town Hall at 2023 Annual Meeting, AGU, 11-15 December.
2. **Preisser, M.**, A. Jha, J. Ali, X. Sui, S. Adelsperger, (2023), Building Your Network: Sharpening the Soft Skills of Science, TH43B Co-Convened Town Hall at 2023 Annual Meeting, AGU, 11-15 December.
1. Patterson, J., L. Miralha, **M. Preisser**, (2022), Building Your Network: AGU-H3S Networking, TH13D Co-Convened Town Hall at 2022 Annual Meeting, AGU, 12-16 December.

ADDITIONAL WORK EXPERIENCE

GIS Analyst Intern <i>HDR Engineering</i>	Mar. 2019 – Aug. 2019 <i>Austin, Texas</i>
Sustainability Senior Student Associate <i>University of Texas at Austin Office of Sustainability</i>	Aug. 2018 – May 2019 <i>Austin, Texas</i>
ReNUWIT REU Program <i>Colorado School of Mines: Advised by Dr. Tzahi Cath</i>	May 2017 – July 2017 <i>Golden, Colorado</i>
DoE SULI Intern <i>Energy & Environment. Directorate, Pacific Northwest National Lab</i>	May 2016 – July 2016 <i>Seattle, Washington</i>
Engagement and Outreach Intern <i>Auburn University Office of Sustainability</i>	Aug. 2015 – May 2018 <i>Auburn, Alabama</i>